

THE PROBLEM

≈1,150

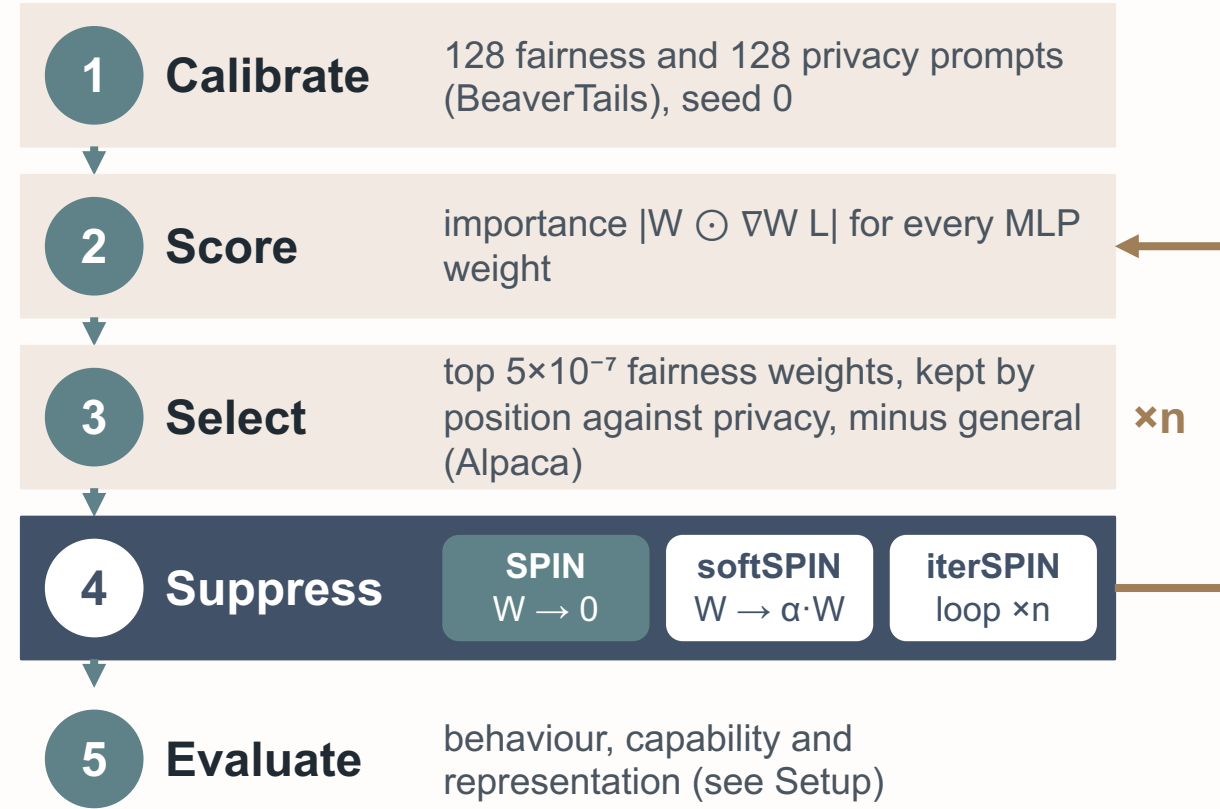
weights SPIN zeroes, out of 5.7 billion MLP weights in Qwen2-7B

SPIN's claim: a tiny set of weights carries both refusals, fairness and privacy.

A clinical LLM must refuse biased requests and requests that expose patient data, and training for privacy can make a model less fair (Qian et al., 2025). SPIN aims to zero the weights the two share: no retraining, both refusal rates rise. **But it is a switch, not a dial: one cut, found once, tested on general text only.**

THE METHOD

One pipeline, two changes



Training-free; $\alpha = 1$ is the untouched model, $\alpha = 0$ and $n = 1$ are SPIN. Step 3 follows the released code: indices paired by position.

THE SETUP, AND WHY

Models, data, three kinds of measure

Models	gemma-2-2b, Qwen2.5-3B, Llama-3.2-3B; scale-up Qwen2-7B, SPIN's own model, so results compare directly
Behaviour	MD-Judge safe rate. General: SALAD-Bench fairness $n = 2,165$, privacy $n = 657$. Clinical: EquityMedQA $n = 1,416$, MedSafetyBench $n = 657$
Capability	WikiText-2 and medical perplexity; MMLU; MedMCQA and MMLU clinical
Representation	HSIC over 4 kernel bandwidths; normalised HSIC and linear CKA (clinical)
Controls	untouched model; original SPIN; top-k cuts of the same size as iterSPIN
Stats	paired exact McNemar; one seed; ≈138 GPU-hours on Kaggle 2× T4 node
Replication	Qian et al. +8.1 / +10.4 pts; this work +5.3 / +6.5 from a higher baseline (both $p < 10^{-4}$)

RQ1 softSPIN

Gap: all-or-nothing zeroing $\rightarrow$ scale the weights by α

QUESTION

Can scaling ($W \rightarrow \alpha W$) match zeroing for less capability cost?

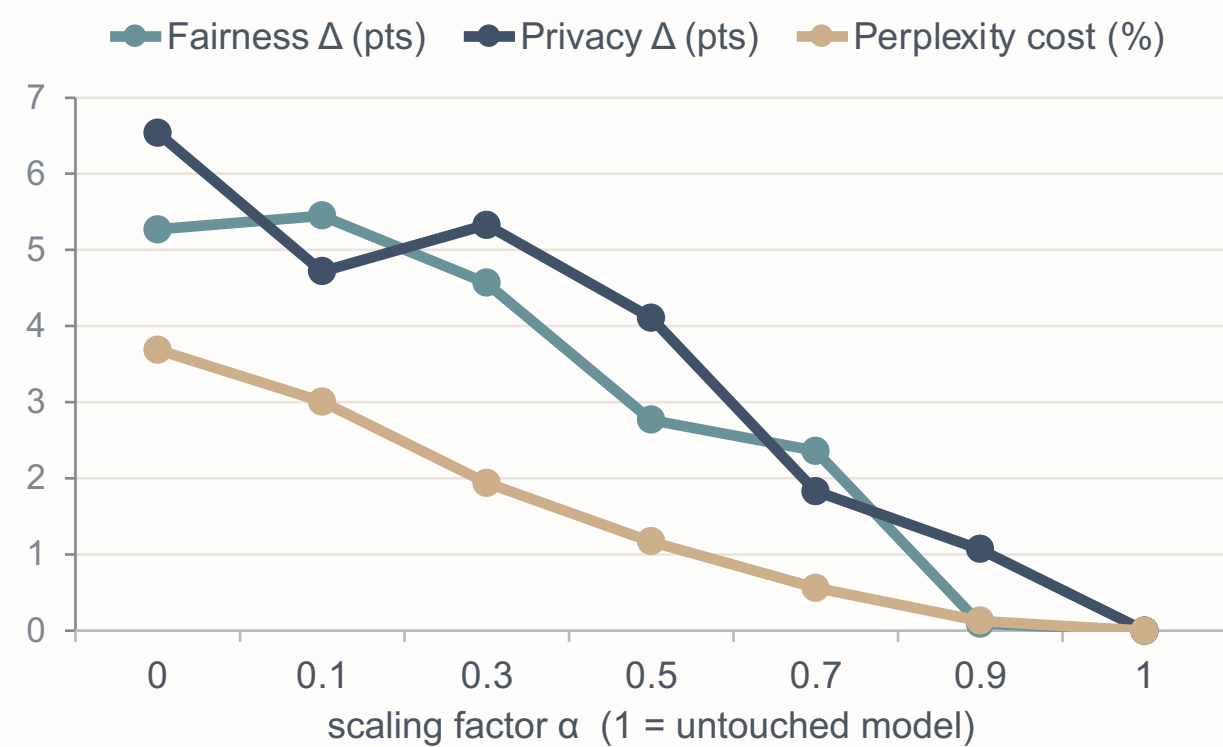
MAIN RESULT · QWEN2-7B

+5.3 / +6.5

$\alpha = 0$ (SPIN): fair / priv pts, +3.7% perplexity

+4.6 / +5.3

$\alpha = 0.3$: fair / priv pts, +1.9% perplexity



At $\alpha = 0.3$ both gains sit about 1 pt below SPIN, the size of the run-to-run noise, for half the perplexity cost.

ACROSS MODELS · SIGN DEPENDS ON MODEL

Model	$\alpha = 0$ fair	$\alpha = 0$ priv	$\alpha = 0.3$ fair	$\alpha = 0.3$ priv	Δ PPL $\alpha = 0$
Qwen2-7B	+5.27	+6.54	+4.57	+5.33	+3.7%
Qwen2.5-3B	+11.27	-6.24	+8.27	-4.72	+14.8%
Llama-3.2-3B	-2.40	0.00	-0.60	+0.46	-0.4%
gemma-2-2b	+0.32	+1.52	+0.37	-0.61	+1.7%

Safe-rate change in points; grey = not significant. A trade-off on Qwen2.5-3B, harmful on Llama, inert on gemma.

Answer to RQ1: yes, on Qwen2-7B: a light setting keeps most of the gain (81-87%) at half the cost. It is a per-model dial, not a general improvement.

RQ2 iterSPIN

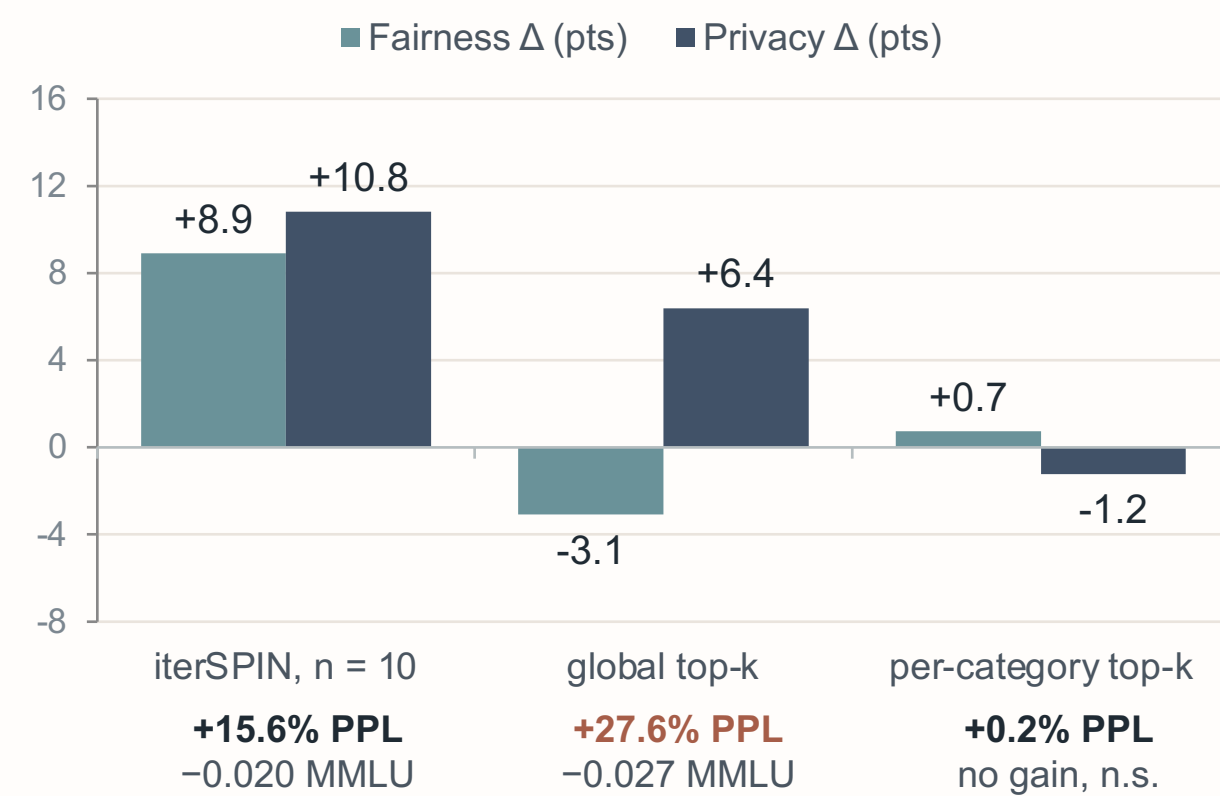
Gap: weights found in one pass $\rightarrow$ re-find them over n rounds

QUESTION

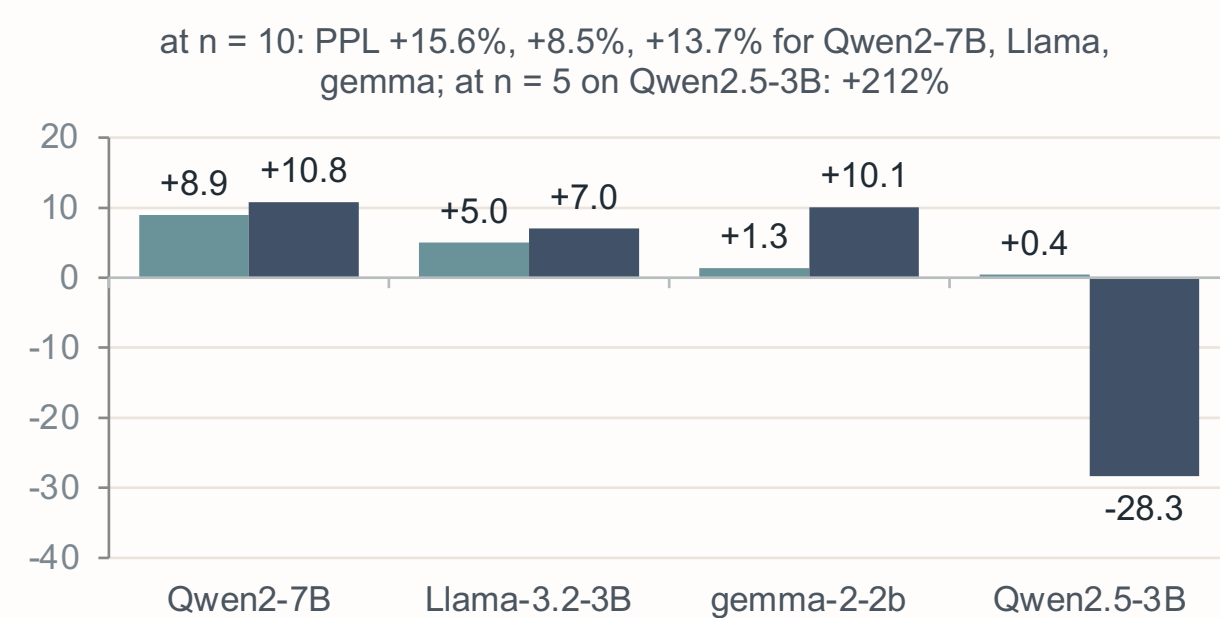
Does re-finding the weights on the edited model, over n rounds, find a better set?

MAIN RESULT · SAME BUDGET, DIFFERENT CHOICE

Three cuts of Qwen2-7B, each removing ≈8,590 weights. Only re-identification improves both axes.



ACROSS MODELS · HELPS THREE, BREAKS ONE



Why Qwen2.5-3B breaks: on the other models the newly picked weights fall by a fifth to a half over 10 rounds as the set settles; here only 7%. The score stops finding a coherent set: privacy -28.3 pts, MMLU -11.4 pts.

Answer to RQ2: yes: a more effective set, not a sparser one (7.5× SPIN's footprint), and only while the selection keeps settling.

RQ3 The clinical case study

Gap: tested on general text only $\rightarrow$ clinical case study

QUESTION

Does the coupling persist on clinical data, and still help both axes?

THE COUPLING, AND WHAT IS ZEROED

1,145

true intersection (1,877 general)

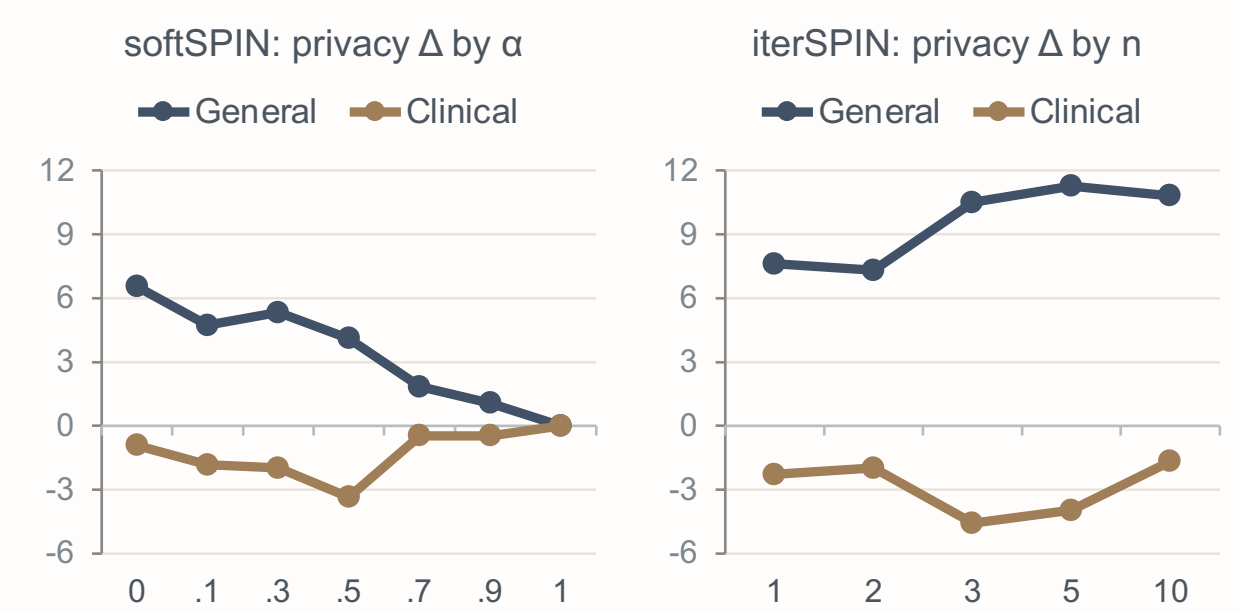
≥66%

of zeroed weights are fairness-only

-3.4%

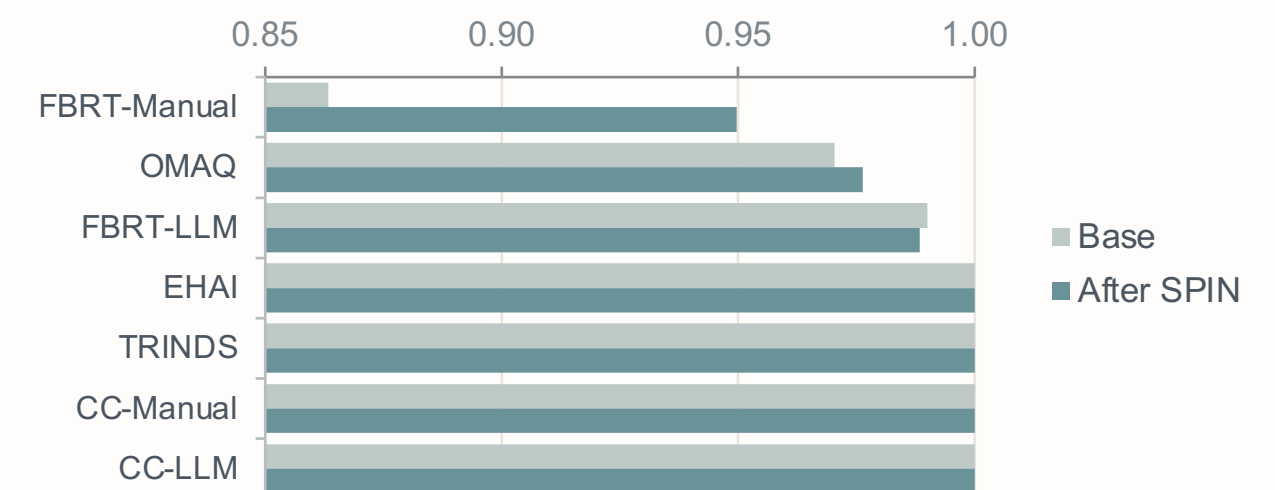
normalised HSIC; CKA -0.78%

MAIN RESULT · PRIVACY FLIPS SIGN



0 of 12 clinical settings improved privacy. The biggest drops (-4.6 at $n = 3$, -3.4 at $\alpha = 0.5$)

FAIRNESS: ONLY WHERE THERE IS ROOM



Pooled +0.85 pts, inside the ~1-pt noise floor; the gain is FBRT-Manual (+8.6). Clinical skill kept: MedMCQA 0.339 $\rightarrow$ 0.343, medical perplexity +2%.

Answer to RQ3: the coupling persists; its privacy benefit does not. Not a drop-in safety tool for clinical use.

LIMITS, AND THE FIX FOR EACH

One run per setting, no multiple-testing correction

Repeat runs of the same cut differ by ~1 pt, so 1–2 pt clinical effects are exploratory.

Fix: 3 seeds on headline settings; Holm per sweep.

SPIN as released, not as specified

The released code selects weights by position, not by intersection, so results describe SPIN as it runs.

Fix: re-run the anchors with a true intersection.

Raw HSIC is confounded for softSPIN

Rescaling shrinks activations, so raw HSIC falls whether or not dependence does.

Fix: normalised HSIC from stored activations.

One behavioural instrument

A single refusal judge (MD-Judge) scores both axes; no human evaluation.

Fix: a second judge on the headline settings.

CONCLUSION

SPIN's fairness–privacy fix is a dial, not a switch. A lighter setting keeps nearly all of the gain at half the cost and choosing the right weights beats removing more. In the clinical case study, the privacy benefit does not transfer.

Contributions C1 softSPIN, with a decoupling argument (Appendix A) · C2 iterSPIN against same-size controls · C3 first SPIN-class clinical study · C4 HSIC's link to behaviour is unstable across models

Also found SPIN's released code selects weights by position, not by intersection: in the clinic, at least two-thirds of the zeroed weights lie outside the privacy set.

Still contested whether the coupling lives in individual weights or is distributed. Editing weights cannot settle it.

References Qian et al. (2025) The Tug of War Within: Mitigating the Fairness-Privacy Conflicts in Large Language Models, ACL · Li et al. (2024) SALAD-Bench, Findings of ACL · Pfohl et al. (2024) A toolbox for surfacing health equity harms and biases in LLMs, Nature Medicine · Han et al. (2024) MedSafetyBench, NeurIPS D&B · Gretton et al. (2005) Measuring statistical dependence with Hilbert-Schmidt norms, ALT · Kornblith et al. (2019) Similarity of neural network representations revisited, ICML

+4.6 / +5.3

fair / priv pts at 1.9% perplexity ($\alpha = 0.3$)

+8.9 / +10.8

fair / priv pts, iterSPIN; same-size cuts fail

0 / 12

clinical settings that improved privacy